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Uncovering thousands of endosymbiont DNA transfer events within single cockroach genomes

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Abstract

Horizontal gene transfer (HGT) between organisms can be a valuable source of genetic variation and innovation. ~~While HGT is prevalent among prokaryotes, it is generally believed to occur infrequently in eukaryotes.~~ Research on HGT in eukaryotes has hitherto focused on transfers of coding sequences; insertions of non-coding DNA remain poorly understood. Here, we investigated HGT in cockroaches, which have a long-standing evolutionary relationship with the ~~transovarially~~vertically transmitted endosymbiont *Blattabacterium cuenoti*, making them a valuable system for assessing the potential scale of HGT. We aligned 150 bp genomic fragments of *Blattabacterium cuenoti* to 23 cockroach and termite genomes, including 8 genomes newly sequenced, and revealed pervasive endosymbiont DNA transfer events. Australian panesthiine ~~and~~ geoscapheine cockroaches were consistently found to harbor ~~>over~~ 3000 HGT inserts, more than an order of magnitude higher than the previous maximum estimate in other eukaryotes, excluding rotifers. Some inserts appear to have persisted for >28.7 million years in panesthiine cockroaches~~this group~~, which may reflect functional roles. We identified numerous chimeric inserts comprising up to nine short segments from different locations in the *Blattabacterium cuenoti* genome. Our findings indicate ~~that HGT may be considerably more prevalent in eukaryotes than previously thought~~pervasive HGT in eukaryote genomes, with potentially far-reaching implications for adaptation and speciation.

Main Text

Introduction

~~Horizontal gene transfer (HGT) challenges conventional Darwinian evolutionary models by introducing genetic material across species boundaries.~~ The number of ~~transferred~~ DNA inserts ~~acquired via horizontal gene transfer (HGT)~~ that become fixed in a species is expected to represent only a small fraction of the ~~actual~~total number of ~~acquisition~~transfer events that occur ~~across~~over evolutionary time [1]. While there is an abundant body of research ~~focused~~ on insertions with homology to protein-coding sequence, transfers that do not encode proteins, or with unknown function, have been largely ignored. Nevertheless, such insertions abruptly introduce genetic variation that can persist if fitness effects are minimal, if purifying selection is insufficient, or if the inserts acquire functional roles [2].

We explored the prevalence and characteristics of DNA inserts originating from the endosymbiont *Blattabacterium cuenoti* (hereafter *Blattabacterium*), in the genomes of 14 cockroaches and 4 termites, including 8 newly sequenced using long-read technology. *Blattabacterium* is found in specialized cells of cockroach fat bodies and is inherited ~~maternally~~transovarially [3]. ~~It infected~~Infecting the ancestors of cockroaches >150–200 Mya [4], ~~but~~Blattabacterium was lost early in termite evolution. Although long-term obligate endosymbionts ~~are thought to have been reported~~ to contribute no or few DNA transfers to their host genomes, previous studies ~~have~~ focused only on protein-coding sequences [5–8].

Results and Discussion

Cockroach genomes contain thousands of Blattabacterium-derived inserts

By focusing on transferred DNA inserts ≥ 50 bp and applying stringent filters to remove artifactual inserts and inserts potentially derived from duplication events (see SI Appendix), we characterized 40,485 *Blattabacterium*-derived inserts across 23 cockroach and termite genomes (93–4900 per genome; Fig. 1A), with a median size of 161 bp (Fig. 1B). With the exception of

rotifers, which have frequent HGT associated with anhydrobiosis [9], the highest number of transfers reported in any eukaryotic species was previously 170 [10]–282 [8]. Inserts were clustered on some contigs, but generally were broadly distributed across host genomes, consistent with many independent HGT events (see Fig. 1C, Dataset S4 and SI Appendix).

~~There are several reasons why~~ Explanations for the high numbers of inserts we may have identified ~~considerably more inserts~~ in cockroaches are as follows. First, we focused on *Blattabacterium*, an obligate endosymbiont whose genome is likely to come in close contact with host nuclear DNA during ~~egg formation~~ oogenesis and zygote development. Second, we analyzed both intra- and intergenic regions of the genome. A recent study did not detect any horizontal transfers from *Blattabacterium* ~~into cockroaches either of the two cockroaches examined or in the termite *Mastotermes darwiniensis*, all of which harbor this symbiont~~ [8]; however, this is likely due to a focus on ~~protein-coding sequence~~ gene regions. Third, while previous studies ~~have~~ filtered for the completeness of inserted coding sequences, we identified inserts as short as 50 bp. Fourth, previous studies often relied on short-read sequencing, where inserts spanning ~~the majority~~ most of a sequence read may have been misidentified as bacterial contamination and discarded. In contrast, we ~~used long-read sequencing for most panesthiine genomes, which enabled us to detect inserts within single sequence reads, flanked on both sides by cockroach host sequence~~ did not actively exclude *Blattabacterium*-like sequences during assembly, and analysed genomes generated using both short- and long-read data (Dataset S2). Long-read data enables detection of inserts flanked by host sequence that may have been missed when using short-read data. Using the same pipeline, we identified comparatively low numbers of inserts from *Wolbachia*, and few to no inserts from *Buchnera* (which infects aphids, and not cockroaches) (see SI Appendix).

Panesthiinae and Geoscaphinae were found to possess particularly high numbers of *Blattabacterium*-derived inserts (Fig. 1A). ~~This~~ Previous work found that this group has experienced ~~considerably more~~ considerable genome-wide relaxed selection ~~than other cockroaches~~ [11]–[10], possibly due to reduced effective population sizes, which ~~could~~ may allow larger numbers of inserts to persist. The low number of inserts identified in termite genomes may reflect the long-term absence of *Blattabacterium*, as well as a considerable reduction in genome size compared with cockroaches.

Evolution of *Blattabacterium*-derived inserts in cockroach genomes

Examination of insert position across the cockroach genomes revealed that 15.38–25.56% were found in intragenic regions, and 74.44–84.62% were found in intergenic regions. This differs somewhat from the proportion of genic regions in the genomes, which was >25% in all but four (range 13.79–33.71%). The average size of inserts (+/- standard deviation) in intragenic regions (199.13 +/- 138.75 bp) was similar to that in intergenic regions (191.70 +/- 126.24 bp), although significantly higher in the former (one-way nested analysis of variance: $F_{1,39424} = 17.96$, $p < 0.01$; Fig. 1B). Insert sequences had GC content (17.9–29.6%) similar to those of *Blattabacterium* genomes (21.3–28.2%), but considerably lower than those of cockroach and termite genomes (34.5–41.1%). There was no apparent bias in the origin of inserts with respect to their position in the *Blattabacterium* genome, which is highly conserved in size, gene content, and gene order across strains (Fig. ~~1C~~ 1D).

Closely related taxa carried similar numbers of inserts, suggesting that many had been inherited vertically (Fig. 1A). We identified putatively ancestral HGT inserts by calculating their similarity to inserts in other species and to the host's specific *Blattabacterium* strain, and subsequently inspecting homology (and common origin) of cockroach genome sequence upstream and downstream from these candidate ancestral inserts ($n = 22$) across multiple taxa (Fig. 2A; see SI Appendix for details). The oldest ancestral inserts identified were inferred to have been present in

the most recent common ancestor of all Panesthiinae and Geoscaphinae, at least ~28.7 Mya [1211].

Transcription and chimeric nature of inserts

We used RNAseq data to investigate the transcription of inserts in *Panesthia t. tegminifera*, *Macropanesthia lithgowae*, and *Panesthia sloanei*. By mapping insert sequences to RNA contigs using BLASTn, we found that 91.42–94.96% of inserts showed no evidence of transcription (although RNA from all life stages and tissues was not assessed). Of the remaining inserts, which were observed in assembled RNA contigs, 26.59–32.97% overlapped with annotated genic regions of the cockroach genomes (1.66–2.28% of total inserts), while the remainder overlapped with intergenic regions (3.38–6.30% of total inserts). A small fraction (0.40–0.65%) of inserts mapped to exon sequences within RNA contigs. (see Dataset S5 for gene details).

During manual BLAST inspections, ~~we noted that several of the longest~~ inserts ~~displayed, we identified 32 inserts displaying~~ a chimeric nature, comprising *Blattabacterium*-like sequences from distant areas within the *Blattabacterium* genome placed directly adjacent to each other within host genome contigs (Fig. 2B). These chimeric inserts were present in all cockroach genomes, comprising up to nine *Blattabacterium*-derived segments ~~from distant genomic regions~~. Such chimeric DNA inserts were present in all cockroach genomes, independently sequenced by separate research groups using different sequencing technologies. Similar to These chimeric inserts are akin to previous reports of chimeric numts previously identified in *Arabidopsis thaliana* [1312], and they are likely to have been produced by non-homologous end-joining repair that incorporated segments of *Blattabacterium* DNA that possibly arose following symbiont cell replication or death.

Conclusion

Our findings suggest that reveal extensive horizontal transfer of DNA from prokaryote symbionts to eukaryotes may be considerably more pervasive than previously thought. Furthermore, the persistence of numerous inserts over millions of years indicates that they may have assumed functional roles in both genes and intergenic regions, are effectively neutral, or are only slightly deleterious. Future research on cockroaches and other species harboring obligate endosymbiotic prokaryotes will help to uncover any functional effects of inserts, providing a more comprehensive understanding of how horizontal transfer HGT shapes genome evolution.

Materials and Methods

Detailed methods are described in the SI Appendix and on GitHub (https://github.com/MEEP-projects/HGT_pipeline). Briefly, we generated 8 new genome assemblies from species of Blaberidae (Dataset S1), and analyzed these with 15 previously generated genomes from Blattodea and 77 *Blattabacterium* (Dataset S2). To locate DNA transfers from *Blattabacterium* to host genomes, we aligned 150-bp fragments of the ~~77~~ *Blattabacterium* genomes to the cockroach and termite genomes. Aligned fragments were characterized as putative HGT inserts, with a minimum length of 50 bp. Several filtering steps and QC analyses were applied to confirm that the identified inserts were 'true' HGT inserts, rather than assembly artifacts or misidentified *Blattabacterium* DNA, and to identify unique insert events within a genome (i.e., excluding inserts arising from duplication or transposition). We also ran control analyses using *Buchnera* (negative control) and *Wolbachia* (known to infect hosts) using the same pipeline. After characterizing inserts, we investigated their minimum age, length, origin, insertion locality, and GC content.

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Figure legends

Figure 1. Dated evolutionary tree of analyzed taxa alongside ~~the~~ total number of HGT inserts and ~~the~~ number of unique inserts (excluding potential cases of duplication or transposition) (A). Ranges are given for taxa with multiple analyzed genomes. Node ages (Mya) are based on previous studies [~~12~~11,13,14,15]. ~~The length~~. Length distribution of inserts (B). 50 bp represents our minimum length threshold, and inserts >1000 bp are omitted for clarity (51 ~~of 39,411~~ inserts ~~omitted~~). ~~The cumulative~~Cumulative number of inserts plotted along host genomes (C). Individual lines correspond to individual contigs (coloured by genome), such that each line starts at position 0 and terminates at the length of the contig. Cumulative number of inserts plotted along the *Blattabacterium* genome (~~CD~~), based on the start position of each insert. Linear trends indicating random origins from the *Blattabacterium* genome. Only species with a published, host-specific *Blattabacterium* strain are included (Dataset S2). For species with multiple ~~published~~ genomes, the version with the most inserts was analyzed (for ~~both 'B' and 'C'~~B-D).

Figure 2. ~~Example of how a putative~~Putative ancestral insert ~~was~~ evaluated by ~~examining the examination of flanking region~~ homology of its flanking regions (A). ~~The right part of panel 'A' shows the homology~~Homology of an insert (green bar) and its 5-kb flanking regions (blue bars) across ~~the species in which it was identified is shown on right~~ (species lacking the insertion are shown with dashed lines ~~in the phylogenetic tree~~). ~~The arrow on left~~. Arrowhead denotes ~~the inferred timing of this ancestral insertion, at the most recent common ancestor of the cockroach genomes carrying the insertion event~~. Example of a chimeric HGT insertion in the *Panesthia tryoni tegminifera* genome (B). The *P. t. tegminifera* contig (top) is divided into seven segments of different origin, and the circular *Blattabacterium cuenoti* genome (CP142624.1; bottom) indicates the source of these insertions. Numbers indicate corresponding segments in the cockroach and *Blattabacterium cuenoti* genomes (with yellow labels marking genomic coordinates).

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